10012 - Mapping water in the Solar System with asteroids in the JWST public archive

Cycle: 5, Proposal Category: AR

INVESTIGATORS

Name	*Institution*
Dr. David E. Trilling (PI)	**Northern Arizona University**
Alicia Allen (CoI)	Northern Arizona University
Dr. John A. Stansberry (CoI)	Space Telescope Science Institute
Dr. Bryan Jason Holler (CoI)	Space Telescope Science Institute
Mr. Bryan Hilbert (CoI)	Space Telescope Science Institute
Ryder Strauss (CoI)	Northern Arizona University
Dr. Cristina A. Thomas (CoI)	Northern Arizona University
Dr. Andrew S. Rivkin (CoI)	The Johns Hopkins University Applied Physics Laboratory

OBSERVATIONS

ABSTRACT

The JWST public archive offers an extraordinarily rich opportunity to map the presence of hydration and, by proxy, water in the asteroid belt, without requiring any new observations. We propose here to harvest serendipitously observed asteroids from the JWST archive to measure the fraction of main belt asteroids that are water-bearing. We will search for water absorption features on 15,000--150,000 asteroids -- the first large-scale survey for water on sub-kilometer asteroids. We will be sensitive to absorption features as small as 10% and provide a new estimate for the amount of water present in the asteroid belt, with implications for planetary system formation and chemistry and the origin of life on Earth. Our expected sensitivity is well matched to LSST, so all our objects, almost all of which will be new discoveries, will eventually also have known orbits, colors, taxa, and lightcurves. Only JWST offers the space-based sensitivity to carry out this project, and an archive program is the most efficient way to complete this project.

OBSERVING DESCRIPTION